

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Practical Lab Exam 2

Practical Lab Exam 2

Class: Class 8

Assessment Type: Practical Lab Exam 2

Total Marks: 20

Duration: 45 minutes

Topics Covered: SQL with Python and Web Development

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Lab Tasks (15 marks)

Task 1: SQL Database Operations with Python (5 marks)

Write a Python program using SQLite that:

- Creates a database `school.db` with a table `Students` containing:
 - `roll_no` (INTEGER PRIMARY KEY)
 - `name` (TEXT NOT NULL)
 - `class` (TEXT)
 - `section` (TEXT)
 - `marks` (INTEGER)
- Inserts at least 8 records
- Performs and displays results of:
 - SELECT with WHERE clause
 - SELECT with ORDER BY
 - UPDATE a student's marks
 - DELETE a student record
 - COUNT and AVG aggregate queries

Task 2: HTML and CSS Web Page (5 marks)

Create a complete HTML page `student_portal.html` with an external CSS file `styles.css` that includes:

- A styled header with school name and navigation links
- A table displaying 6 students with columns: Name, Class, Section, Marks, Grade
- The table should have:
 - Alternating row colours
 - Hover effect on rows
 - Proper borders and spacing
- A styled footer with copyright information
- CSS should use flexbox for layout and responsive design

Task 3: SQL Query Challenge (5 marks)

Given the following database schema:

```
Employees(emp_id, name, department, salary, joining_date)
Projects(project_id, project_name, emp_id, start_date, end_date)
```

Write Python code that:

- Creates both tables with proper relationships (Foreign Key)
- Inserts at least 6 employees and 10 projects
- Executes and displays the results of:

- Find all employees who earn more than the average salary
- List departments with more than 2 employees
- Find employees working on the most projects
- Calculate total salary expenditure per department
- Find projects that are still ongoing (no end date or end date > today)

Viva Voce (5 marks)

1. What is the difference between INNER JOIN and LEFT JOIN? Give one example of each. (1 mark)
2. How does CSS specificity work? Which rule takes priority: inline, internal, or external CSS? (1 mark)
3. What is the purpose of a Foreign Key in SQL? How does it maintain data integrity? (1 mark)
4. What is the difference between WHERE and HAVING in SQL? When would you use each? (1 mark)
5. Explain the difference between responsive design and adaptive design in web development. (1 mark)